

Name: .....

Q(ABCDEG)

$F = \{AB \rightarrow C, AC \rightarrow D, D \rightarrow EG, G \rightarrow B, A \rightarrow D, CG \rightarrow A\}$

Find the minimal cover?

**Step 1:  $F' = F$ , Write all the functional dependencies such that each contains exactly one attribute on its right side**

$F' = \{AB \rightarrow C, AC \rightarrow D, D \rightarrow E, D \rightarrow G, G \rightarrow B, A \rightarrow D, CG \rightarrow A\}$

**Step 2: Check the essentiality of each functional dependency one by one.**

-  $AB \rightarrow C$

+  $AB \rightarrow C, \{AB\}^+ = \{A, B, C, D, E, G\} = R$

+  $AB \rightarrow C, \{AB\}^+ = \{A, B, D, E, G\}$

$\Rightarrow$  the two results are different

-  $AC \rightarrow D$

o  $AC \rightarrow D, \{AC\}^+ = \{A, B, C, D, E, G\} = R$

o  $AC \rightarrow D, \{AC\}^+ = \{A, B, C, D, E, G\} = R$

$\Rightarrow$  the two results are same

$F' = \{AB \rightarrow C, D \rightarrow E, D \rightarrow G, G \rightarrow B, A \rightarrow D, CG \rightarrow A\}$

-  $D \rightarrow E$

o  $D \rightarrow E, \{D\}^+ = \{D, E, G, B\}$

o  $D \rightarrow E, \{D\}^+ = \{D, G, B\}$

$\Rightarrow$  The two results are different.

-  $D \rightarrow G$

o  $D \rightarrow G, \{D\}^+ = \{D, G, B, E\}$

o  $D \rightarrow G, \{D\}^+ = \{D, E\}$

$\Rightarrow$  The two results are different.

-  $G \rightarrow B$

o  $G \rightarrow B, \{G\}^+ = \{G, B\}$

o  $G \rightarrow B, \{G\}^+ = \{G\}$

$\Rightarrow$  The two results are different.

-  $A \rightarrow D$

o  $A \rightarrow D, \{A\}^+ = \{A, B, C, D, E, G\} = R$

o  $A \rightarrow D, \{A\}^+ = \{A\}$

⇒ The two results are different.

-  $CG \rightarrow A$

○  $CG \rightarrow A, \{CG\}^+ = \{A, B, C, D, E, G\} = R$

○  $CG \rightarrow A, \{CG\}^+ = \{C, G, B\}$

⇒ The two results are different.

⇒  $F' = \{AB \rightarrow C, D \rightarrow E, D \rightarrow G, G \rightarrow B, A \rightarrow D, CG \rightarrow A\}$

**Step-03: Consider the functional dependencies having more than one attribute on their left side.**

-  $AB \rightarrow C$

○  $\{AB\}^+ = \{A, B, C, D, E, G\} = R$

○  $\{A\}^+ = \{A, D, G, E, B, C\} = R$

○  $\{B\}^+ = \{B\}$

⇒  $A \rightarrow B \Rightarrow F' = \{A \rightarrow C, D \rightarrow E, D \rightarrow G, G \rightarrow B, A \rightarrow D, CG \rightarrow A\}$

-  $CG \rightarrow A$

○  $\{CG\}^+ = \{A, B, C, D, E, G\} = R$

○  $\{C\}^+ = \{C\}$

○  $\{G\}^+ = \{G, B\}$

⇒  $F' = \{A \rightarrow C, D \rightarrow E, D \rightarrow G, G \rightarrow B, A \rightarrow D, CG \rightarrow A\} \Rightarrow$  the minimal cover